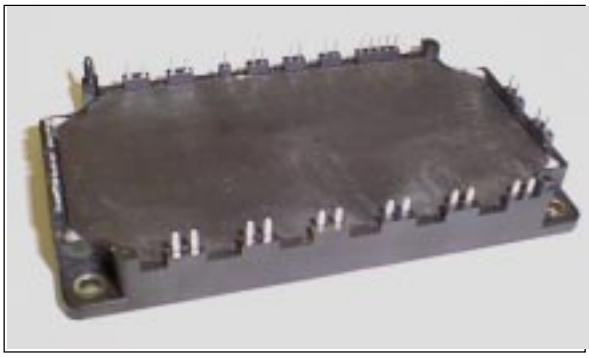
7MBR75SD060

IGBT Modules

PIM/Built-in converter with thyristor and brake (S series)
600V / 75A / PIM



■ Features

· Low V_{CE}(sat)
· Compact Package
· P.C. Board Mount Module
· Converter Diode Bridge Dynamic Brake Circuit

■ Applications

· Inverter for Motor Drive
· AC and DC Servo Drive Amplifier
· Uninterruptible Power Supply

■ Maximum ratings and characteristics

● Absolute maximum ratings (Tc=25°C unless without specified)

Item		Symbol	Condition	Rating	Unit
Inverter	Collector-Emitter voltage	V_{CES}		600	V
	Gate-Emitter voltage	V_{GES}		±20	V
	Collector current	I_C	Continuous	75	A
		I_{CP}	1ms	150	A
		$-I_C$		75	A
	Collector power disspation	P_C	1 device	300	W
Brake	Collector-Emitter voltage	V_{CES}		600	V
	Gate-Emitter voltage	V_{GES}		±20	V
	Collector current	I_C	Continuous	50	A
		I_{CP}	1ms	100	A
	Collector power disspation	P_C	1 device	200	W
	Repetitive peak reverse voltage(Diode)	V_{RRM}		600	V
Thyristor	Repetitive peak off-state voltage	V_{DRM}		800	V
	Repetitive peak reverse voltage	V_{RRM}		800	V
	Average on-state current	$I_{T(AV)}$	50Hz/60Hz sine wave	75	A
	Surge 0n-state current (Non-Repetitive)	I_{TSM}	Tj=125°C, 10ms half sine wave	750	A
	Junction temperature	T_{jw}		125	℃
Converter	Repetitive peak reverse voltage	V_{RRM}		800	V
	Average output current	I_O	50Hz/60Hz sine wave	75	A
	Surge current (Non-Repetitive)	I_{FSM}	Tj=150°C, 10ms	525	A
	I^2t (Non-Repetitive)	I^2t	half sine wave	1378	A^2s
Junction temperature (except Thyristor)		T_j		+150	℃
Storage temperature		T_{stg}		-40 to +125	℃
Isolation between terminal and copper base *2		V_{iso}	AC : 1 minute	AC 2500	V
voltage between thermistor and others *3				AC 2500	V
Mounting screw torque				1.7 *1	N·m

*1 Recommendable value : 1.3 to 1.7 N·m (M4)
*2 All terminals should be connected together when isolation test will be done.
*3 Terminal 8 and 9 should be connected together. Terminal 1 to 7 and 10 to 26 should be connected together and shorted to copper base.

● Electrical characteristics (Tj=25°C unless otherwise specified)

Item		Symbol	Condition		Characteristics Min.	Typ.	Max.	Unit
Inverter	Zero gate voltage collector current	I_{CES}	V_{CE}=600V, V_{GE}=0V				200	μA
	Gate-Emitter leakage current	I_{GES}	V_{CE}=0V, V_{GE}=±20V				200	nA
	Gate-Emitter threshold voltage	$V_{GE(th)}$	V_{CE}=20V, I_C=75mA		5.5	7.8	8.5	V
	Collector-Emitter saturation voltage	$V_{CE(sat)}$	V_{GE}=15V, Ic=75A	chip		1.8		V
				terminal		2.1	2.55	
	Input capacitance	C_{ies}	V_{GE}=0V, V_{CE}=10V, f=1MHz			7500		pF
	Turn-on time	t_{on}	V_{CC}=300V			0.45	1.2	μs
		t_r	I_C=75A			0.25	0.6	
	Turn-off	t_{off}	V_{GE}=±15V			0.40	1.0	
		t_f	R_G=33Ω			0.05	0.35	
	Forward on voltage	V_F	I_F=75A	chip		1.7		V
				terminal		2.0	2.7	
	Reverse recovery time of FRD	t_{rr}	I_F=75A				300	ns
Brake	Zero gate voltage collector current	I_{CES}	V_{CES}=600V, V_{GE}=0V				200	μA
	Gate-Emitter leakage current	I_{GES}	V_{CE}=0V, V_{GE}=±20V				200	nA
	Collector-Emitter saturation voltage	$V_{CE(sat)}$	I_C=50A, V_{GE}=15V	chip		1.8		V
				terminal		2.05	2.55	
	Turn-on time	t_{on}	V_{CC}=300V			0.45	1.2	μs
		t_r	I_C=50A			0.25	0.6	
	Turn-off time	t_{off}	V_{GE}=±15V			0.40	1.0	
		t_f	R_G=51Ω			0.05	0.35	
	Reverse current	I_{RRM}	V_R=600V				200	μA
	off-state current	I_{DM}	V_{DM}=800V				1.0	mA
Thyristor	Reverse current	I_{RRM}	V_{RM}=800V				1.0	mA
	Gate trigger current	I_{GT}	V_D=6V, I_T=1A				100	mA
	Gate trigger voltage	V_{GT}	V_D=6V, I_T=1A				2.5	V
	On-state voltage	V_{TM}	I_{TM}=75A	chip		1.1	1.18	V
				terminal		1.2		
Converter	Forward on voltage	V_{FM}	I_F=75A	chip		1.1		V
				terminal		1.2	1.5	
	Reverse current	I_{RRM}	V_R=800V				200	μA
Thyristor	Resistance	R	T=25°C			5000		Ω
			T=100°C		465	495	520	
	B value	B	T=25/50°C		3305	3375	3450	K

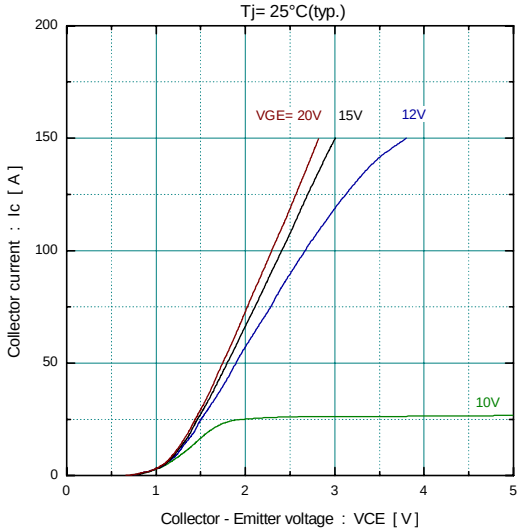
● Thermal resistance Characteristics

Item	Symbol	Condition	Characteristics Min.	Typ.	Max.	Unit
Thermal resistance (1 device)	Rth(j-c)	Inverter IGBT			0.42	°C/W
		Inverter FWD			0.90	
		Brake IGBT			0.63	
		Thyristor			0.56	
		Converter Diode			0.70	
Contact thermal resistance *	Rth(c-f)	With thermal compound		0.05		

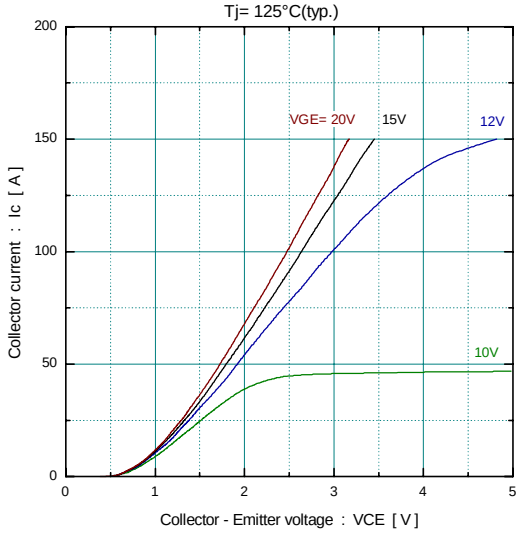
* This is the value which is defined mounting on the additional cooling fin with thermal compound

■ Characteristics (Representative)

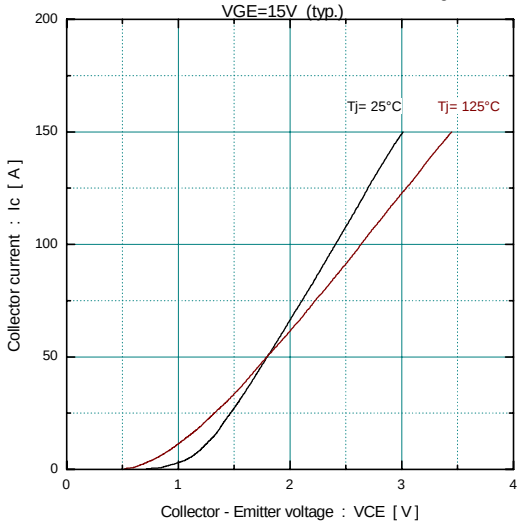
[Inverter]
Collector current vs. Collector-Emitter voltage
Tj= 25°C(typ.)



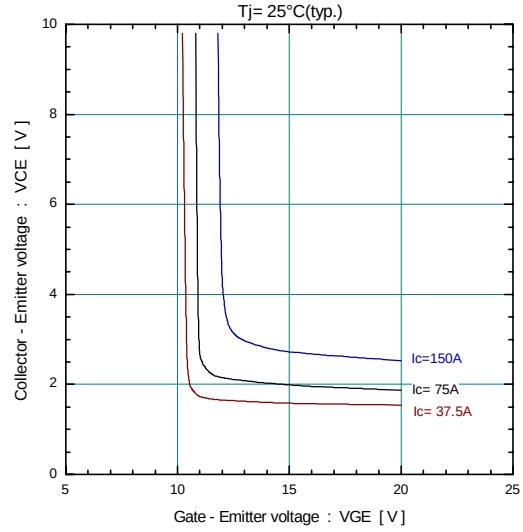
[Inverter]
Collector current vs. Collector-Emitter voltage
Tj= 125°C(typ.)



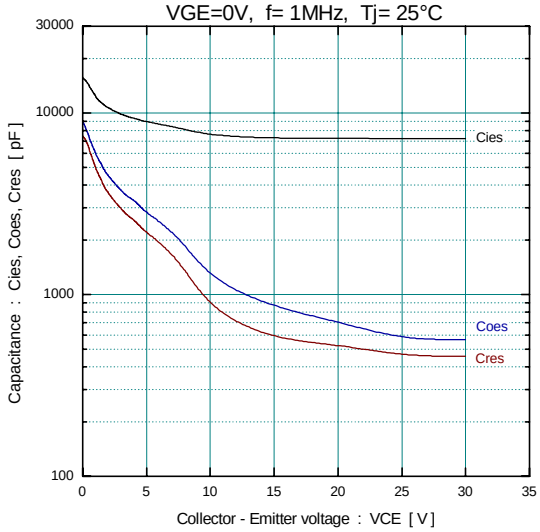
[Inverter]
Collector current vs. Collector-Emitter voltage
VGE=15V (typ.)



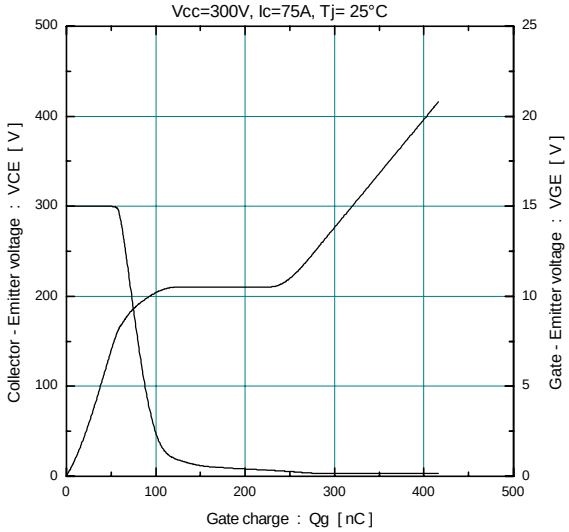
[Inverter]
Collector-Emitter voltage vs. Gate-Emitter voltage
Tj= 25°C(typ.)



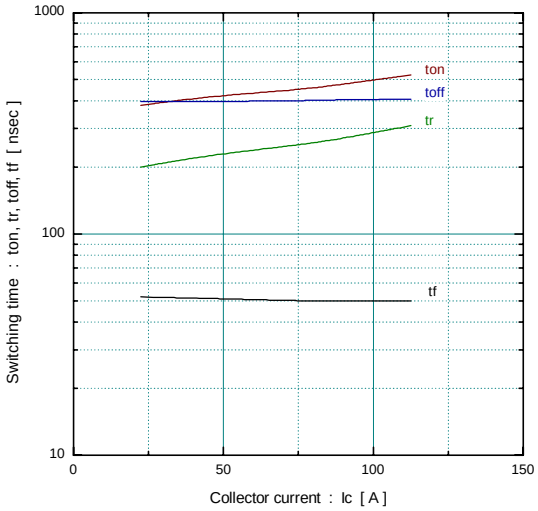
[Inverter]
Capacitance vs. Collector-Emitter voltage (typ.)
VGE=0V, f= 1MHz, Tj= 25°C



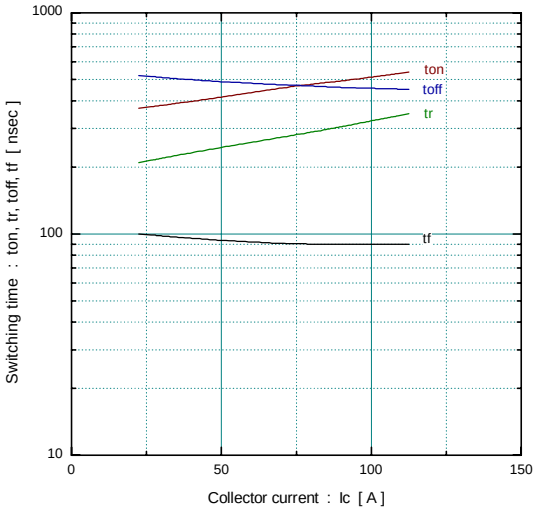
[Inverter]
Dynamic Gate charge (typ.)
Vcc=300V, Ic=75A, Tj= 25°C



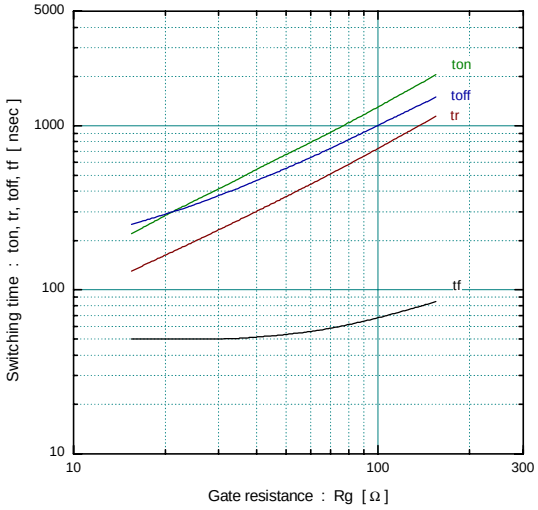
[Inverter]
Switching time vs. Collector current (typ.)
Vcc=300V, VGE=±15V, Rg=33Ω, Tj= 25°C



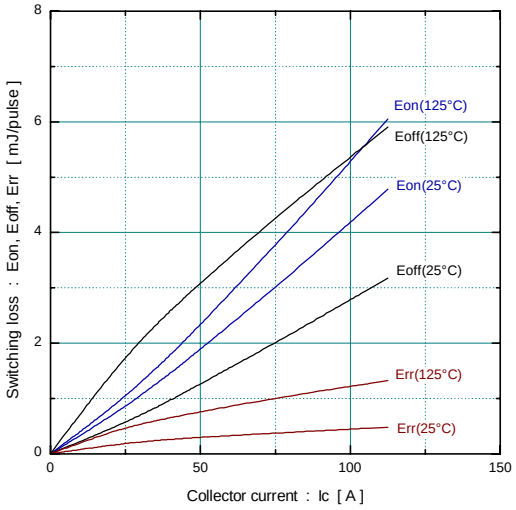
[Inverter]
Switching time vs. Collector current (typ.)
Vcc=300V, VGE=±15V, Rg= 33Ω, Tj= 125°C



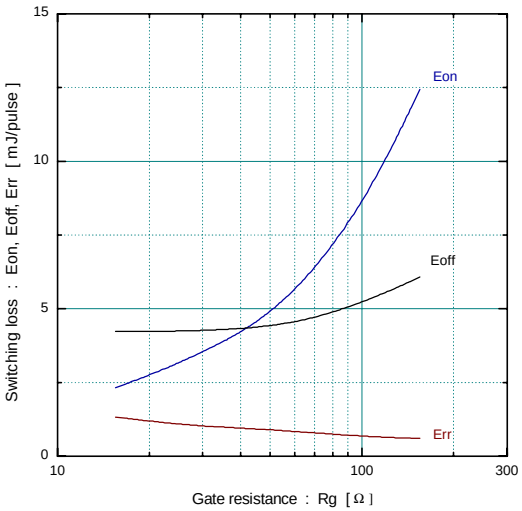
[Inverter]
Switching time vs. Gate resistance (typ.)
Vcc=300V, Ic=75A, VGE=±15V, Tj= 25°C



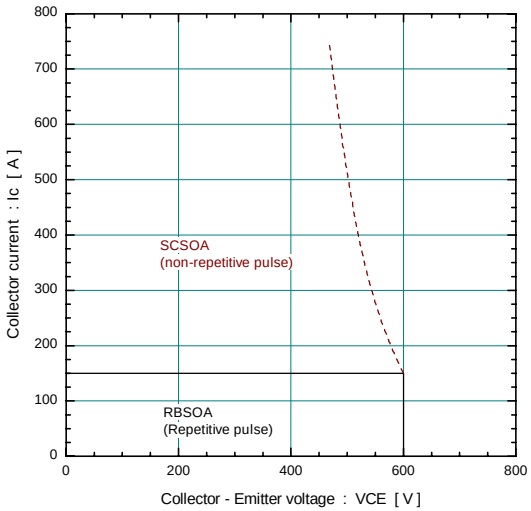
[Inverter]
Switching loss vs. Collector current (typ.)
Vcc=300V, VGE=±15V, Rg=33Ω



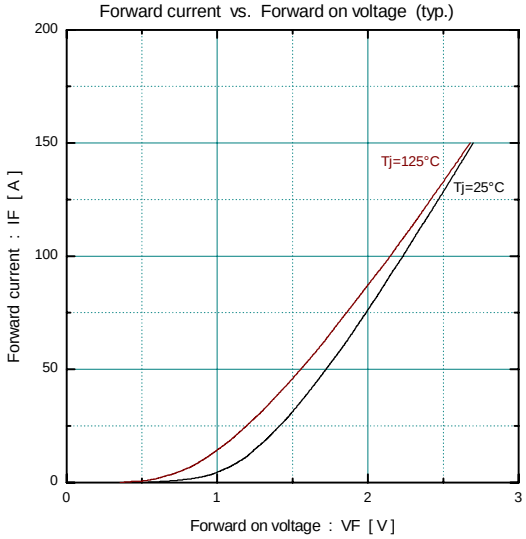
[Inverter]
Switching loss vs. Gate resistance (typ.)
Vcc=300V, Ic=75A, VGE=±15V, Tj= 125°C



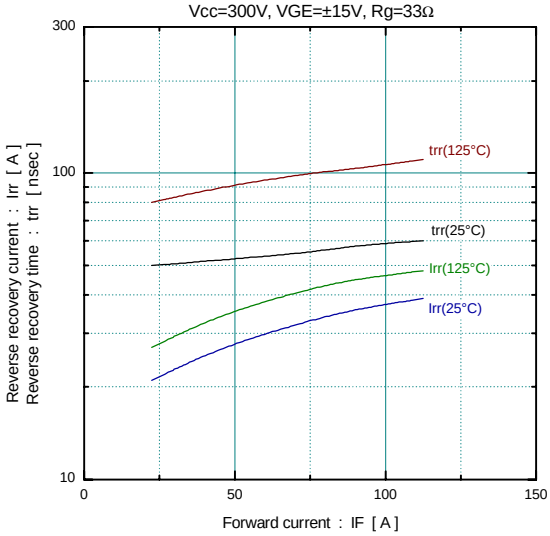
[Inverter]
Reverse bias safe operating area
+VGE=15V, -VGE<=15V, Rg>=33Ω, Tj<=125°C



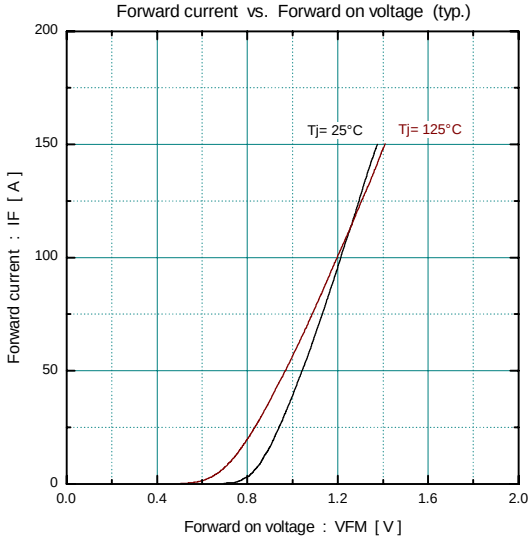
[Inverter]
Forward current vs. Forward on voltage (typ.)



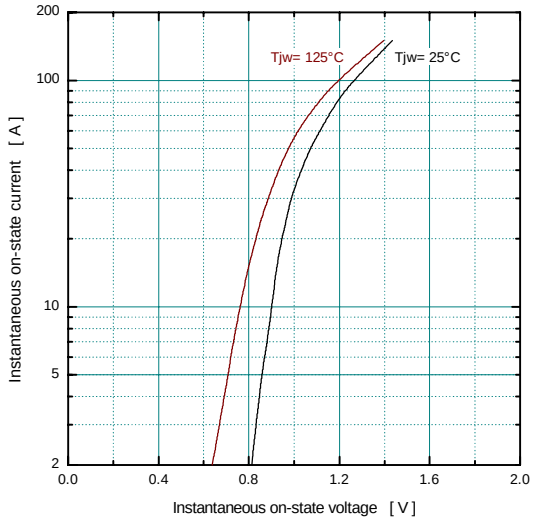
[Inverter]
Reverse recovery characteristics (typ.)
Vcc=300V, VGE=±15V, Rg=33Ω



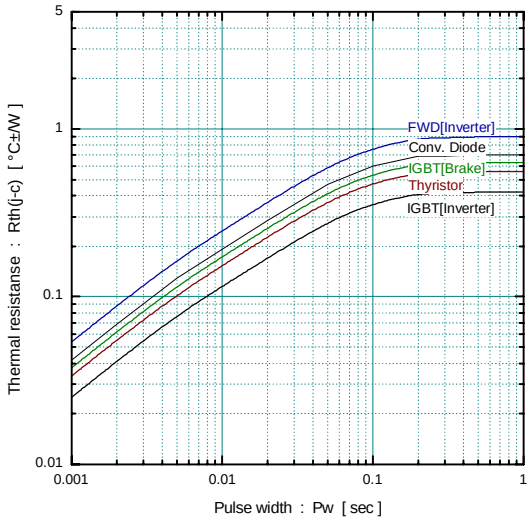
[Converter]
Forward current vs. Forward on voltage (typ.)



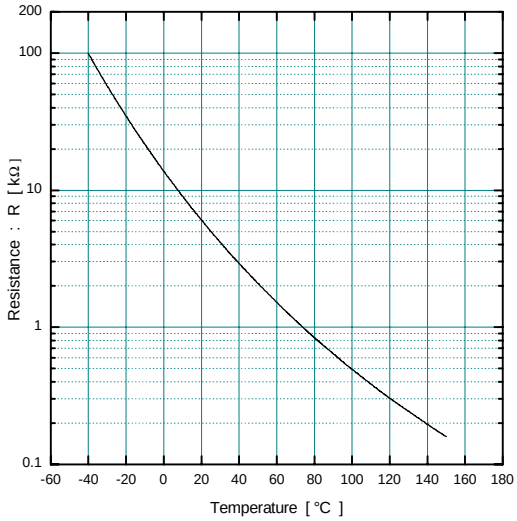
[Thyristor]
On-state current vs. On-state voltage (typ.)



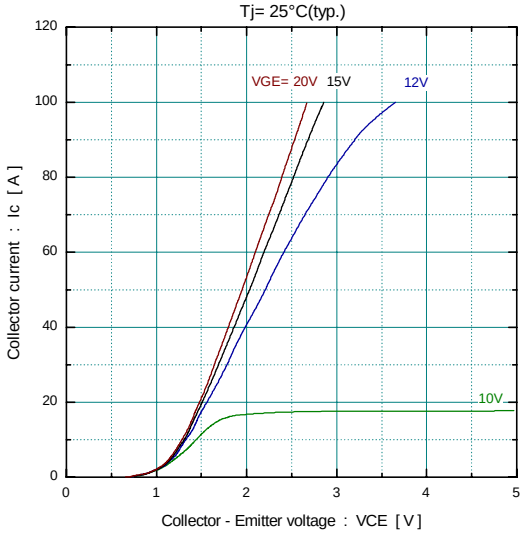
Transient thermal resistance



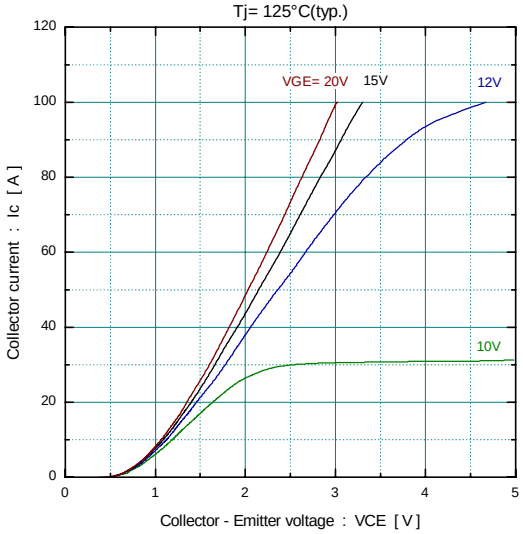
[Thermistor]
Temperature characteristic (typ.)



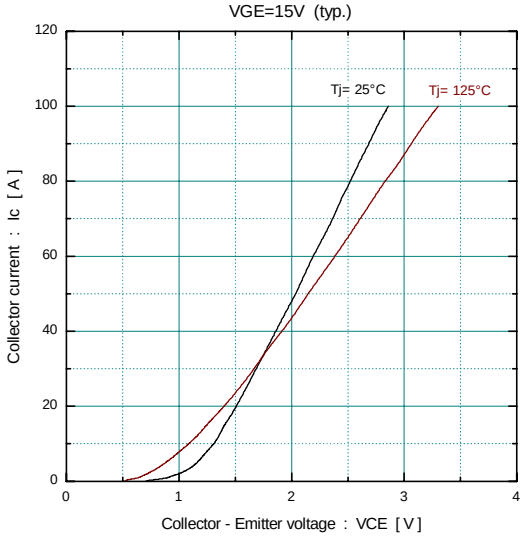
[Brake]
Collector current vs. Collector-Emitter voltage
Tj= 25°C(typ.)



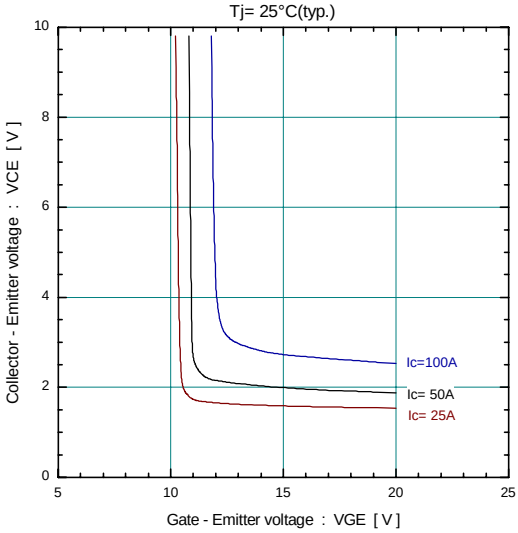
[Brake]
Collector current vs. Collector-Emitter voltage
Tj= 125°C(typ.)



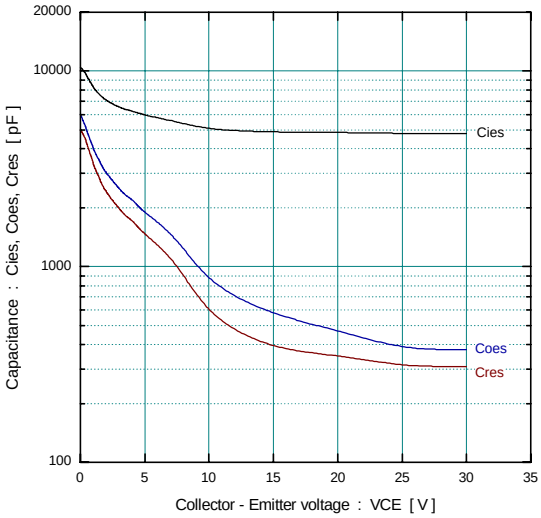
[Brake]
Collector current vs. Collector-Emitter voltage
VGE=15V (typ.)



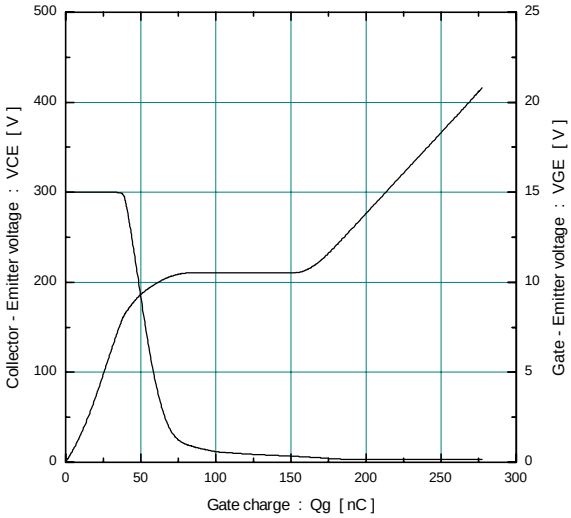
[Brake]
Collector-Emitter voltage vs. Gate-Emitter voltage
Tj= 25°C(typ.)



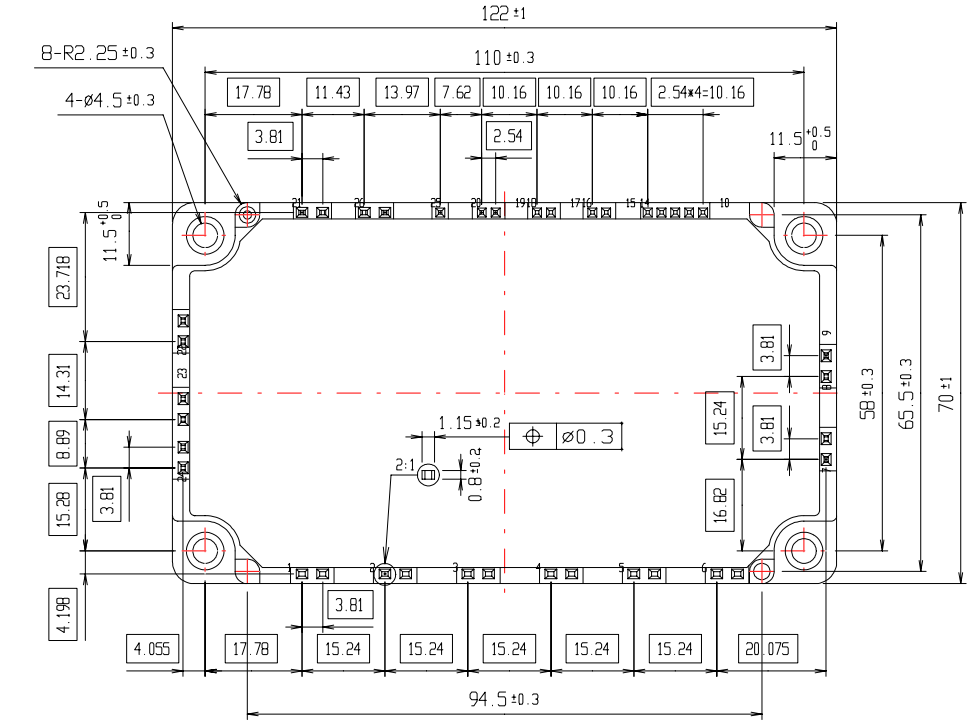
[Brake]
Capacitance vs. Collector-Emitter voltage (typ.)
VGE=0V, f= 1MHz, Tj= 25°C

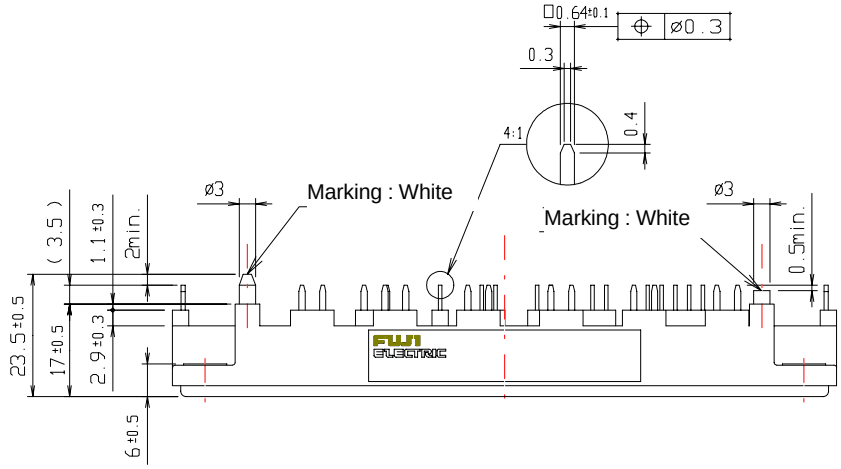


[Brake]
Dynamic Gate charge (typ.)
Vcc=300V, Ic=50A, Tj= 25°C



■ Outline Drawings, mm





■ Equivalent Circuit Schematic

